

Smart City Traffic Analytics: Project Report

1. Project Overview

This project analyses interstate highway traffic data containing **48,187 hourly records** to understand the patterns of how the traffic builds up, and the reasons for the different variations. A clear, step by step pipeline was built using Python:

- The raw traffic data and weather numbers are prepared and cleaned.
- Practical features to understand traffic trends is created.
- Generated Visual charts explaining commuter habits and effects of weather.
- Created a terminal application where users can look at the statistics, and filter traffic by weather or time of the day

2. Data preparation methodology

Three main steps applied to change the raw numbers to traffic insights which can be used for analysis

Use of cyclical encoding to make time work on a 24 hour

If standard numbers are used the 11pm (2300 hrs) and midnight (0000 hours) look very far apart even though its just an hour apart in real life.

To fix this, we used basic trigonometry ($\sin$ and $\cos$) to place hours on a round 24-hour clock. This allows the data to be understood that midnight naturally follows late night without a huge difference mathematically.

Levelling the Playing Field (Min-Max Scaling)

The traffic volume ranges between 0-7000 cars/hour, while the range of temperature varies across the everyday outdoor values.

This has to be adjusted, without which large number can significantly overpower smaller numbers when calculations are done. Temperature was converted to Celsius, and both temperature and traffic volume was scaled to fit between 0 and 1, so that there is a balanced view

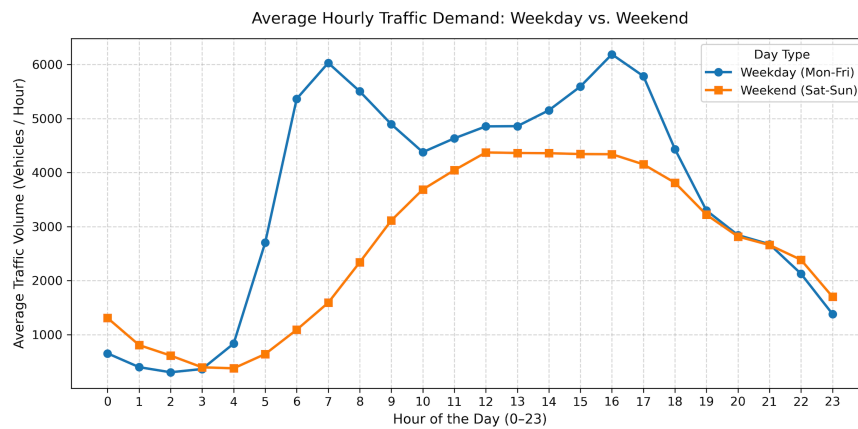
Dividing Traffic into 4 Real-World Categories (Quartile Buckets)

The actual traffic data (48,187 records) was divided into 4 equal parts (25% chunk) to categorise the data from Severe traffic to Low traffic.

- Low : 0 to 1,192 vehicles/hour
- Medium : 1,193 to 3,379 vehicles/hour
- High : 3,380 to 4,933 vehicles/hour
- Severe : 4,934 to 7,280 vehicles/hour

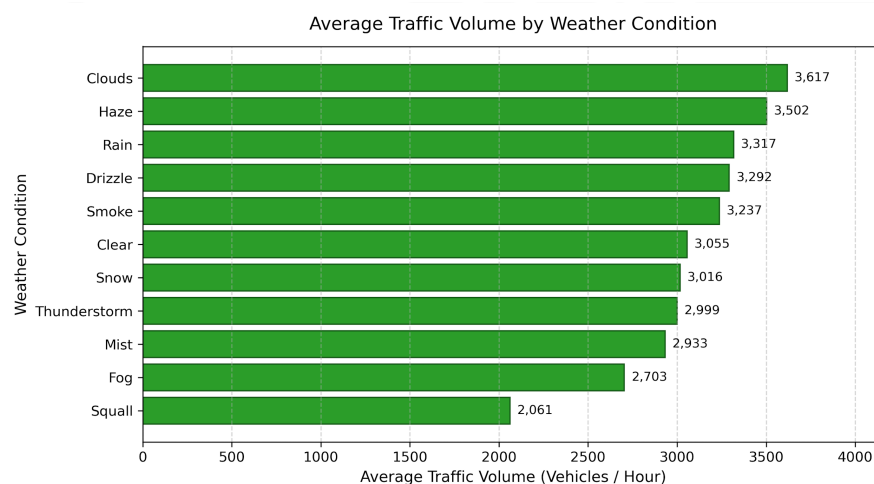
3. Conclusions from the visual insight diagrams

Chart 1: Weekday vs. Weekend Traffic patterns



- Two big traffic spikes on weekdays is clearly shown. Morning peak time between 6 am to 8 am (approximately 6000 cars per hour) and evening peak time between 3 pm to 5 pm (approximately 6000 cars per hour)
- During the weekend, the traffic flow doesn't have any peak time spikes. It indicates a smooth traffic building up gradually during the morning, and stays relatively steady and flat across mid day, and then drops off in the evening.
- Both weekdays and weekends have the quietest traffic between 2:00 am to 4:00 am (well under 1000 cars/hr), although from midnight to 2am, the traffic is slightly higher on weekends compared to weekdays, probably because of weekend outings.

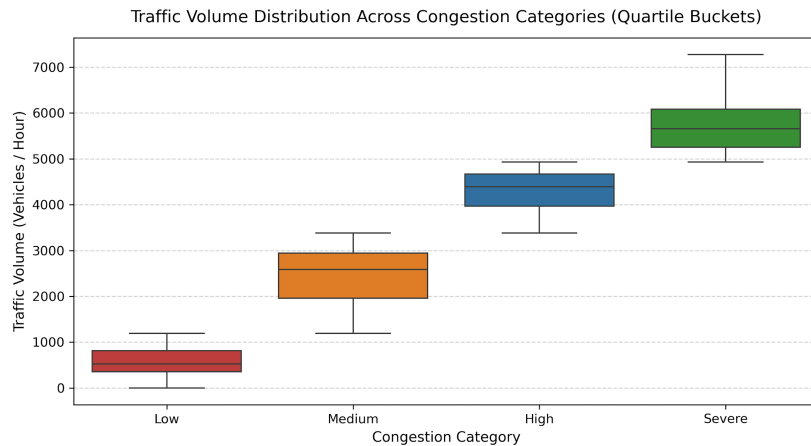
Chart 2: Impact of Weather on Roads



- The traffic volume drops considerably during severe weather conditions. As seen, during Squalls which creates dangerous conditions, the volume was the lowest (~ 2061 vehicles/hour), followed by Fog (~ 2703 vehicles/hour).
- During normal weather changes, there is very little changes. Normal daily commutes continue to happen regardless of minor changes in weather.

(cloudy weather (~ 3617 vehicles/hour), light haze (~ 3502 vehicles/hour), and rain (~ 3317 vehicles/hour)

Chart 3: Clear categorisation of Congestion Levels



- The congestion boxplot shows very clear four categories with no overlaps between the range of each category.
- The largest range is seen in the Severe category which ranges from ~4900 to ~ 7300 vehicles/hour) indicating the predictability issues and bottlenecks happen during peak time.

4. Take Aways for City planners

1. During weekday, the roads are heavily congested between 6 am to 8 am and 3 pm to 5 pm. Introducing higher toll charges during these peak times, and providing lesser or no tolls during the non peak hours can encourage people travelling for non urgent or discretionary trips to a quieter period
2. When sudden changes in weather happens especially in severe weather scenarios, highway signs should immediately alert drivers to reduce speed, and also introduce more highway support teams to attend and assist the commuters.